Examiner reports

Q1.

This question was answered well by most students. Many scored 2 out of a possible 3 marks and could find the inverse function and state it in a correct form. The most frequent mistake seen was incorrectly stating the domain or missing it out altogether. The mark for the correct domain was independent and could be achieved without correctly finding $f^{-1}(x)$, as it could be deduced from the range of $f(x)$.

Q2.

This question discriminated well between students of varying abilities. The majority of students understood that they had to integrate and chose the appropriate technique, often applying integration by parts successfully. A significant number of students made no attempt to go any further and completely ignored the constant of integration. These students generally achieved three marks.

Students who realised they had to determine the constant of integration often made good progress and scored 7 or 8 marks. Those who dropped marks at this point generally showed insufficient detail to fully justify their answer. Several very thorough, complete solutions were seen.

Q4.

In part (a), the full algebraic expression $f(x) \geq -\frac{4}{3}$ was expected here. But the most common error was the reversal of the inequality, with $f(x) \leq -\frac{4}{3}$ seen on many scripts.

In part (b)(i), most students followed through their value from part (a), although many lost the mark for incorrect notation by using $f(x)$ rather than x. Some students failed to see the relationship between the answers to parts (a) and (b)(i) and wrote down inequalities such as $x \geq 0$ or $x \leq 0$.

In part (b)(ii), almost all students were able earn 2 marks by reversing the operations correctly and replacing y with x. But even the strongest students had difficulty gaining full marks by giving the answer $-\sqrt{3x+4}$. There were some students who interpreted the inverse function to mean $\frac{3}{x^2-4}$.

Part (c)(i) was answered very well. However, some students who reached the equation $3x = 2$ gave the solution as $x = \frac{3}{2}$. Other errors seen were $3x - 1 = 0$ and $3x - 1 = e$.

In part (c)(ii), most students gained both marks. However, some students wrote “No” without an adequate reason; they needed to say “because it is not one to one” or “because it is many to one”. Statements about there being a modulus or a In or an asymptote were not relevant.

Part (c)(iii) was generally answered well. A small number of students made errors in cancelling or in adding –4 and –1. Very few students attempted to find fg(x) rather than gf (x) .

Part (c)(iv) was very poorly answered. Most students only earned the method mark for writing $x^2 - 5 = 1$. The students who considered $x^2 - 5 = -1$ usually gained the first accuracy mark. Very few students gave all four roots and a fully correct solution by rejecting the positive roots.

Q5.

This question was tackled well with many students gaining 11 marks. It was good to see a better understanding of the connection between the exponential and logarithmic functions than in previous series.

(a) (i) This part was well answered with many fully correct responses. The great majority of the students gained the mark for swapping x and y; most students did so at the start. Marks were lost by a small number of students because they could not change logarithm to exponential form.

(ii) To earn the mark it was essential to give $f^{-1}(x) > \frac{3}{2}$, but many used x or y or f^{-1} or range or ≥ and these were not acceptable.

(iii) The correct graph was usually seen and many students gained full marks. However, some students allowed the gradient in the second quadrant to become negative, some had the wrong intercept and some had the graph in completely the wrong place, failing to recognise they needed to reflect in the line $y = x$.

(b) (i) The correct composition was usually applied and many students went straight to the answer. There were some students who had difficulty with manipulating $e^{2\ln(2x-3)}$ to $e^{\ln(2x-3)2}$.

(ii) Most students chose the correct composition and gained the method mark. Although many students went on to complete the question correctly, handling of the logarithms proved beyond many and $\ln(2e^{2x} - 11)$ was often changed into $4x - 11$. Necessary brackets were often omitted.

Q6.

In part (a), there were many fully correct answers, but often the accuracy mark was lost for two separate sets not connected. Some candidates gave their answers as strict inequalities, and some gave their answers as x instead of f(x) and a few gave the range to be 21 – 1 = 20. However many weaker candidates did not know how to tackle this part at all.

In part (b)(i), it was good to see that most got a correct expression, although a few candidates wrote $\dfrac{\frac{63}{x}+1}{4}$ as $\dfrac{252}{x}+\dfrac{1}{4}$. Some candidates made a sign error, getting –1 instead of +1. It was good to see that hardly anyone left their expression in terms of y or took the function to be $\dfrac{4x-1}{63}$.

Part (b)(ii) was very well answered with most candidates obtaining both marks. Those candidates who had made an error in part (b)(i) were usually able to obtain the method mark for a correct step.

Part (c)(i) was, again, mostly well done with just a few having $\dfrac{63}{(4x)^2}-1$ or $\dfrac{63}{(4x-1)^2}$.

In part (c)(ii), very few candidates, even the most able, gained full marks. Most candidates gained the first two marks by equating to 1 and obtaining an equivalent expression to x^2 = 16, but most then offered the two solutions +4 and –4 and failed to state that the only possible solution was –4, thus taking no account of the domain for g.

Q7.

Although this question was well answered very few students scored full marks. The majority of students scored full marks in parts (a) and (b), although there was a common mistake in part (b) of $\dfrac{20}{x}$ = 25 – 5. Even the better students failed to obtain both marks in part (c)(ii), with few scripts giving a justification of why there was only one solution and rejecting x = –3.

Q8.

This question was generally answered very well and full marks were often seen.

In part (a) many correct answers with correct notation were seen but there were many cases of 3 ≤ f(x) ≤ –3 also seen. Where candidates lost a mark it was usually for poor notation.

In part (b)(i) most candidates earned the method mark for swapping x and y but the $\dfrac{1}{2}$ confused

many when trying to obtain $\cos^{-1}\dfrac{x}{3}$, with $\cos^{-1}\dfrac{2x}{3}$ and $\cos^{-1}\dfrac{x}{6}$ being common errors.

In (b)(ii) having the correct inverse function generally led to the correct answer here; unfortunately repeating incorrect algebra for part (b)(i) sometimes gave the 'right' answer, but obviously without reward.

Part (c)(i) was very well answered.

In part (c)(ii) most candidates achieved the method mark by drawing at least two continuous parts. Unfortunately many lost the final mark, as drawing multiple curves was a common error, as was labelling the x-axis incorrectly.

Part (d) was well done; a few candidates had $\frac{1}{3}$ or $\frac{1}{2}$ as the scale factor and a handful introduced another wrong transformation.

Q9.

In part (a) candidates should be advised to answer this question as “not 1 to 1”. For those who chose this phrasing, it was not always clear whether they were referring to f or its inverse when they said that it was “many to one” or perhaps “one to many”. There were many imprecise statements about square roots.

Part (b) was generally well done with only the odd algebraic error. However those who failed to swap x and y at the end were penalised, and those who did this as a first step tended to score better.

In part (c) too often the negative sign was omitted.

In (d) it was good to see fg(x) almost always correct. Many correct approaches also gave $x = -\frac{1}{2}$ as an answer. Equating and then trying to square root both sides, or taking both terms to the same side seldom proved fruitful. A few took $(2x + 1)^2$ as $4x^2 + 1$ which was disappointing at this level, and $2x^2 + 4x + 1$ was also quite common.

Q10.

Considerably less than half the candidates gained 2 marks in part (a); f(x) > –2 was common, as was f(x) > 3.

Part (b)(i) was generally well answered with many fully correct responses seen. The majority of candidates earned the mark for swapping x and y. Marks were lost in the attempt to isolate x or y because many candidates could not cope with changing $e^{2x} = y + 3$ into $2x = \ln (y + 3)$, the most common error being $\ln y + \ln 3$.

In part (b)(ii), the majority of candidates who had been successful in part (b)(i) and knew that $e^0 = 1$ went on to earn both marks. There did, however, seem to be a significant number of candidates who did not know that $e^0 = 1$.

Part (c)(i) was well answered by most candidates.

Part (c)(ii) was reasonably well answered by the majority of candidates, with many earning full marks. Candidates who had trouble with e^{2x} in part (b)(i) also had the same problems in this part.

Q11.

(a) (i) The first part of this question was reasonably answered with many candidates obtaining both marks. Where candidates obtained 1 mark it was because many ended up with the answer $1/(5x – 2)$.

(ii) Again this part was very well answered with candidates obtaining both marks.

The majority of candidates arrived at $K\cos 2x$ but $K = -2$ was a common error.

(b) (i) This part was not answered very well. Many candidates lost a mark through using –0.693 instead of ln ($\frac{1}{2}$) and f(x) ≥ 0 was a common response.

(ii) Most candidates were able to do this part with the correct answers often seen. The main error was the omission of brackets around $5x - 2$ obtaining sin2(ln5x – 2). The expression for fg(x) was also often seen.

(iii) For those candidates with a correct starting expression many went on to get full marks. Those candidates who used sin ln $(5x - 2)^2$ often lost an accuracy mark for not rejecting one of their answers. Most candidates obtained the first method mark for making the correct initial step for their expression.

(iv) This was usually well done but there were common errors of dividing by sin2 obtaining y/sin2 = x or even y/sin = 2x and y/2 = sin$x$

Q12.

Part (a) was reasonably well answered although many candidates again lost a mark through poor notation.

In part (b), candidates usually gave a correct response that f(x) was a many–one relationship or that it was not a one-to-one relationship. Few numerical examples were seen and there were many responses which simply stated it was because it was x^4.

Part (c)(i) was usually correctly answered, with the majority of candidates evaluating fg(x) in the correct order. Surprisingly few totally correct responses to part (c)(ii) were seen, with the majority of incorrect results coming from candidates not realising that $(x - 4)^4 = \ldots$ had both a positive and a negative solution. Consequently, those candidates who were able to invert correctly often only obtained one solution.

Q13.

Part (a) was not very well answered by the majority of candidates. Errors occurred due to poor notation.

Part (b)(i) was very well answered, with most candidates achieving full marks.

Part (b)(ii), like part (a), was not very well answered, with poor notation.

Part (c)(i) was well answered, although candidates often spoiled their work with incorrect subsequent working which was then penalised in the next part.

Many totally correct responses were seen in part (c)(ii) and those candidates who worked with an incorrect h(x) often achieved the method mark for squaring. A common error was to substitute $x = 3$ into h(x).

Q14.

Part (a) was not very well answered with many candidates putting x = **R**. “$x > 0$” was also common.

Part (b)(i) was very well answered.

Part (b)(ii) was answered well by the majority of candidates. Errors were made by those candidates who produced further working in part (b)(i), and hence tried to work with expressions such as $\frac{1}{x^3 - 27}$. Other common incorrect responses were seen, such as x = 7 through mishandling of the 3.

Part (c)(i) was very well answered by the majority of candidates, although $\frac{1}{x} - 3$ was a common incorrect response, which lost the accuracy mark.

Part (c)(ii) was quite well answered with many correct solutions seen. Common incorrect responses were f(x) ≠ 0 and f(x) ≠ –3.

Q15.

Part(a) was fairly well answered but, for many candidates, putting x = □ and $x \geq 0$ was also common.

Part (b)(i) was very well answered. Most candidates at least obtained the 2 method marks but several lost the accuracy mark for an incorrect sign in the numerator. A few candidates tried using a flow chart but these were generally unsuccessful. Part (b)(ii) was answered well by the majority of candidates. Common errors were responses of 0 or 2/3.

Part (c) was not very well answered by the majority of candidates, although most gained part marks. Many candidates only gave the result from the positive square root of 9 or 1/9. Those candidates who formed a quadratic and solved it were far more successful in obtaining both results for the final accuracy mark.

Q16.

Parts (a), (b)(ii) and (c)(ii) proved to be beyond the capability of most candidates. Candidates with correct values tended to struggle with the appropriate notation. In part (c)(i) the majority of candidates found gf, with very few finding fg, however many candidates lost marks for trying unsuccessfully to simplify their answers. There was a poor response to the final part of the question, with being rarely seen.

Q17.

Part (a) was the least successful part of this question. A common incorrect answer was f(x) ≥ 2; other errors often involved poor notation. Many totally correct responses were also seen.

The majority of candidates answered part (b)(i) correctly. The main error seen was $\mathrm{fg}(x) = \frac{1}{\sqrt{x-2}}$. Part (b)(ii) was very well answered by the majority of candidates. Candidates who made errors in part (b)(i) were able to achieve at least the method mark. Part (c) was very well answered by the majority of the candidates.

Q18.

(a) Well answered by the majority of candidates. The most common response by far was $x \geq 0$ which was not penalised on this occasion.

(b) Many candidates obtained full marks. However, a large number of candidates only found one solution because when they took the square root they neglected the negative answer.

(c) This was reasonably well answered although some candidates failed to articulate their answer satisfactorily. Part (ii) was very well answered by the majority of candidates although some lost the final accuracy mark as they ended up with $y = 1/x + 2$.

Q19.

Part (a) This was very poorly answered by the majority of candidates. Range is a topic that is not very well understood by candidates.

Part (b) This was reasonably well answered. Errors occurred where candidates, on interchanging x and y, tried to work with $x = 2e^{3x} - 1$.

Part (c) This was poorly answered by most of the candidates. Many candidates obtained the correct differentiation of the In function but lost marks because they omitted to multiply by 1/2. This resulted in the very common incorrect answer of 2/3.

Q20.

Part (a) was reasonably well answered although some scripts showed evidence of poor algebraic skills when trying to tidy up the expression. Candidates could still score marks on this part even with an incorrect answer to the previous part. Scripts indicated that the basic method was understood although the algebraic skills left a lot to be desired; only a small number of candidates were able to correctly answer the final part.